Variance Decomposition Is a Measurement, Not a Fact: On the Instability of Reproducibility Budgets for LLM Evaluation

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Abstract

We set out to build reproducibility budgets for LLM evaluation using Generalizability Theory (G-theory) and discovered that the budget itself is unreliable. In a multi-benchmark G-study crossing infrastructure (numerical precision, hardware) × methodology (prompt, seed, temperature, ordering) × measurement (item) factors across four models and five benchmarks (172,800–230,400 observations per experiment), we find three results that challenge current variance-attribution practices. First, item dominance estimates are sample-unstable: resampling 200 MMLU items with different stratified seeds yields item variance shares ranging from 25% to 48% (CV = 0.33), meaning prior single-decomposition studies—including all existing G-theory analyses of LLM evaluation—over-commit to attribution certainty. Second, treating model as fixed (the standard practice) systematically inflates item variance from 37.8% to an apparent dominance; a model-as-facet reanalysis reveals that model×item interaction (33%) is the single largest component, reducing pure item variance to 15.2%. Third, D-study reproducibility budgets converge across benchmarks (range = 60 items for $G \geq 0.90$) but carry bootstrap confidence intervals of ±190 items, rendering benchmark-specific budget recommendations statistically indistinguishable and demanding that budgets themselves report uncertainty. These findings reframe variance decomposition from a factual attribution to a measurement with its own error, requiring uncertainty quantification before actionable conclusions can be drawn.

1 Introduction

When LLM evaluation results fail to replicate, the natural response is to identify the culprit: was it the prompt, the random seed, the GPU, the numerical precision, or the item selection? A growing body of work has pursued this question by decomposing evaluation variance into attributable sources [Bouthillier et al., 2021, Mesting, 2026, Brittlebench, 2026]. These studies report point estimates of variance shares—"item selection accounts for 43.8% of variance" [Mesting, 2026], "prompt sensitivity explains 50% of score variation" [Brittlebench, 2026]—and derive practical recommendations from them.

We conducted a large-scale G-theory study with the same goal: cross infrastructure and methodology factors in a unified framework, measure variance components across five benchmarks, and produce reproducibility budgets telling practitioners how many items, prompts, and seeds they need. What we found instead is that *the variance decomposition itself is unstable*. Three independent item samples of 200 MMLU questions yield item dominance estimates of 48%, 25%, and 41%—a coefficient of variation of 33%. Which items happen to be sampled determines not just the magnitude but the *rank order* of variance components: in one sample, the residual exceeds item variance as the largest component. Every prior study that reports a single variance decomposition without resampling stability analysis has, in effect, presented one draw from a high-variance distribution as a finding.

This instability compounds with a second problem: model specification. Standard practice treats model as fixed and pools across models or analyzes each separately. When we instead treat model as a crossed random facet, item variance drops from 37.8% to 15.2%, with the difference absorbed by a model×item interaction of 33%—the largest single component in the crossed design. Item "difficulty" is not an intrinsic property of items but a joint property of item–model pairs, a conclusion that fundamentally challenges the premise of item-dominated variance attribution.

A third discovery concerns reproducibility budgets themselves. D-study projections for $G \geq 0.90$ converge across benchmarks (420–520 items, range = 60), which would seem to simplify practical recommendations. But the bootstrap 95% confidence intervals span ±190 items—wider than the cross-benchmark range—meaning any specific budget recommendation is statistically indistinguishable from its neighbors. Reproducibility budgets require their own error bars.

Our contributions are:

1. Evidence that variance component estimates in LLM evaluation are sample-unstable (item resampling CV = 0.33, component rank inversions across samples), establishing that single-run decompositions over-commit to attribution certainty.

2. A model-as-facet reanalysis showing that model×item interaction (33%) is the largest overlooked variance source, and that treating model as fixed inflates item dominance by >22 percentage points.

3. A demonstration that D-study reproducibility budgets require uncertainty quantification: cross-benchmark budget convergence (range = 60 items) is dominated by within-benchmark CI width (±190 items), rendering benchmark-specific recommendations unreliable.

2 Related Work

Variance decomposition in ML evaluation. Bouthillier et al. [2021] introduced factorial variance decomposition to ML benchmarking via ANOVA, decomposing performance variation across hyperparameters, data splits, and random seeds. Henderson et al. [2018] applied similar methods to deep reinforcement learning, finding that implementation details dominate reported improvements. Dodge et al. [2020] and Sellam et al. [2022] documented fine-tuning instability from seed and data order in BERT-scale models. These studies establish the paradigm we build on but do not address LLM inference evaluation or apply G-theory's D-study projections.

G-theory in LLM evaluation. Mesting [2026] adapted Generalizability Theory to LLM-as-judge evaluation, decomposing variance across five methodological facets. Their study treats model and infrastructure as fixed, reports single-run point estimates without resampling stability analysis, and covers one benchmark. Brittlebench [2026] quantified prompt sensitivity at up to 50% of score variation on fixed hardware. Neither study crosses infrastructure with methodology or tests the stability of their decomposition.

Item-level analysis and IRT. Item Response Theory has been applied to NLP evaluation for item analysis [Lalor et al., 2019, Rodriguez et al., 2021, Vania et al., 2021], efficient subsampling [Polo et al., 2024, Perlitz et al., 2023], and the Benchmark Lottery phenomenon [Dehghani et al., 2021]. These approaches model item properties but do not decompose evaluation variance into infrastructure vs. methodology sources.

Reproducibility factors. Yuan et al. [2025] documented up to 9% accuracy variation from hardware and numerical precision without factorial attribution. Lu et al. [2022], Pezeshkpour & Hruschka [2024] studied option ordering effects in multiple-choice evaluation. Biderman et al. [2024] surveyed evaluation reproducibility challenges. Each isolates at most two factors, implicitly assuming separability.

3 Method

3.1 Generalizability Theory Framework

Generalizability Theory (G-theory; Shavelson & Webb, 1991, Brennan, 2001) decomposes observed score variance into components attributable to facets (e.g., items, prompts, seeds) and their interactions. For a fully crossed design with k facets, a person's observed score X_p is:

$$X_{pijk\ldots} = \mu + \alpha_p + \beta_i + \gamma_j + \delta_k + (\alpha\beta)_{pi} + (\alpha\gamma)_{pj} + \cdots + \epsilon_{pijk\ldots} \tag{1}$$

where each Greek letter represents the effect of a facet or interaction and ϵ is residual error. The G-coefficient summarizes measurement dependability:

$$G = \frac{\sigma_p^2}{\sigma_p^2 + \sigma^2_{\text{relative error}}} \tag{2}$$

D-studies project G under hypothetical facet sample sizes, enabling reproducibility budget planning.

3.2 Experimental Design

We conduct a series of experiments with increasing scope. All experiments use generation-based evaluation (following Hua et al., 2025) with vLLM 0.9.2 and `enforce_eager` mode to eliminate engine-level confounds.

Experiment 1: Full-scale precision study. 4 models (Qwen3-8B, Llama-3.1-8B, Gemma-2-9B, Mistral-7B) $\times$ 3 precisions (FP32, BF16, FP16) $\times$ 6 prompts $\times$ 6 seeds $\times$ 3 temperatures $\times$ 4 orderings $\times$ 200 MMLU items = 172,800 observations. Model is initially treated as fixed (analyzed per-model and pooled); Experiment 4 reanalyzes with model as a crossed facet.

Experiment 1b: Item resampling stability. 3 independent stratified 200-item MMLU samples (seeds 0, 7, 99) $\times$ Llama-3.1-8B $\times$ 3 precisions $\times$ 3 prompts $\times$ 3 seeds $\times$ 2 orderings = 54 conditions per sample, 10,800 observations per sample.

Experiment 2: Multi-benchmark replication. 4 models $\times$ 2 precisions $\times$ 6 prompts $\times$ 6 seeds $\times$ 2 temperatures $\times$ 4 orderings on GSM8K, ARC-Challenge, and HellaSwag (200 items each), 76,800 observations per benchmark.

Experiment 3: Multi-hardware G-study. 4 models $\times$ 2 GPU types $\times$ 2 precisions $\times$ 6 prompts $\times$ 6 seeds $\times$ 3 temperatures $\times$ 4 orderings $\times$ 200 MMLU items = 230,400 observations.

Experiment 4: Comprehensive reanalysis. Merged data from Experiments 1–3. Model-as-facet reanalysis, 1,000-replicate bootstrap CIs on all variance components, D-study projections with uncertainty quantification, and REML sensitivity analysis.

Table 1: Variance components under model-as-fixed analysis (MMLU, binary scoring, 200 items). These point estimates are subsequently shown to be unstable (Sec. 4.2) and inflated (Sec. 4.5).

Component	Qwen3-8B	Llama-3.1-8B	Gemma-2-9B	Mistral-7B	Pooled
item_id	42.2%	35.8%	40.1%	30.3%	37.8%
residual	17.4%	21.2%	18.8%	26.1%	20.2%
temperature:item	6.3%	5.9%	5.2%	5.8%	5.8%
prompt:item	5.8%	5.4%	4.8%	4.2%	5.2%
precision:item	4.9%	4.6%	4.2%	3.8%	4.5%
seed:item	1.8%	1.7%	1.4%	1.3%	1.6%
ordering:item	1.4%	1.2%	1.1%	0.9%	1.2%
precision	1.8%	1.7%	1.5%	1.3%	1.6%
temperature	1.2%	1.0%	0.9%	0.8%	1.0%
prompt	0.8%	0.7%	0.6%	0.5%	0.7%
prec.:temp.	1.6%	1.5%	1.4%	1.2%	1.5%
all other	14.8%	19.3%	20.0%	23.8%	18.9%

Experiment 5: Scale extension. Qwen3-14B and DeepSeek-R1-Distill-14B at BF16/FP16, testing whether 14B models alter variance structure.

Experiment 6: Fine-tuning G-study. RoBERTa-large on SST-2, MNLI, RTE, COPA. 6 seeds $\times$ 3 learning rates $\times$ 4 data shuffles $\times$ 3 GPUs $\times$ 2 precisions = 432 runs per task, 1,728 total.

Experiment 7: LayerCast precision mitigation. FP16 storage + FP32 compute as a fourth precision condition on MMLU.

Experiment 8: MATH benchmark extension. 200 MATH items under the same multi-benchmark protocol, completing a 5-benchmark spectrum.

4 Experiments and Results

4.1 Experiment 1: Variance Decomposition Under Model-as-Fixed Analysis

Table 1 presents the variance decomposition from the full-scale MMLU study with model treated as fixed. At face value, item selection is the largest component (37.8% pooled), with substantial variation across models: Qwen3-8B shows the highest item dominance (42.2%) while Mistral-7B shows the lowest (30.3%, CV across models = 0.13).

Precision main effects are small after confound correction (1.6% pooled), and accuracy differences across FP32 (0.668), BF16 (0.672), and FP16 (0.670) are negligible ($\Delta_{\max} = 0.004$ for each model). The G-coefficient under binary scoring is 0.32, reflecting the inherent low reliability of binary item scoring with 200 items. The precision$\times$temperature interaction (1.5%) is non-negligible but does not by itself break separability.

These results appear to support an item-dominance narrative. However, as we show next, this narrative is an artifact of two methodological choices: a single item sample and a fixed-model specification.

Table 2: Variance decomposition across three independent item samples (Llama-3.1-8B, MMLU). Item dominance is sample-dependent, with rank inversions across samples.

Component	Sample 0	Sample 7	Sample 99	Mean	SD	CV
item_id	48.0%	25.0%	41.0%	38.0%	12.0%	0.33
residual	16.0%	30.0%	19.0%	21.7%	7.5%	0.35
temperature:item	7.2%	9.8%	8.1%	8.4%	1.3%	0.16
prompt:item	5.6%	8.2%	6.4%	6.7%	1.3%	0.20
precision:item	4.8%	6.4%	5.2%	5.5%	0.8%	0.15
temperature	1.0%	2.4%	1.4%	1.6%	0.7%	0.46
prompt	0.6%	1.8%	0.9%	1.1%	0.6%	0.58
all other	16.8%	16.4%	18.0%	17.1%	0.8%	0.05

Table 3: G-coefficient and rank stability across item samples. The rank order of variance components is not stable: Sample 7 reverses the top two components.

Metric	Sample 0	Sample 7	Sample 99
G (correct)	0.28	0.38	0.31
Largest component	item_id	residual	item_id
2nd largest	residual	item_id	residual
3rd largest	temp:item	temp:item	temp:item
Item difficulty SD	0.31	0.22	0.27
Bimodality index	0.71	0.38	0.58

4.2 Experiment 1b: Item Resampling Instability

Table 2 presents the variance decomposition under three independent stratified 200-item MMLU samples. The item variance share ranges from 25.0% (Sample 7) to 48.0% (Sample 0), yielding a coefficient of variation of 0.33. This instability is not merely quantitative: in Sample 7, the residual (30.0%) exceeds item variance (25.0%) as the largest component, producing a qualitatively different decomposition.

Table 3 shows that the G-coefficient itself varies from 0.28 to 0.38 across samples (CV = 0.16), and the largest-component identity is not stable. The bootstrap 95% CI on item variance is [0.27, 0.48] (width = 0.21), wide enough that it cannot distinguish between "item dominance" (>40%) and "item is one of several comparable sources" (∼25%).

The mechanism is traceable: Sample 0 draws more items with extreme difficulty (bimodality index = 0.71), which naturally produce high item-level variance; Sample 7 draws items with more uniform difficulty (bimodality index = 0.38), compressing item variance. MMLU's 57 subjects have heterogeneous difficulty distributions—STEM subjects tend bimodal, humanities more uniform—and 200-item stratified sampling balances at the subject level but not within subjects.

This result is the centerpiece of the paper. Prior work reports single variance decompositions as findings [Mesting, 2026, Brittlebench, 2026, Bouthillier et al., 2021]. A CV of 33% means that if Mesting [2026] had used different items, her 43.8% judge-dominance estimate might have been 30% or 55%. Variance decomposition is a measurement with its own measurement error, and that error has been systematically ignored.

Table 4: Item dominance across benchmarks (model-as-fixed). No systematic gradient exists; spread = 9pp.

Benchmark	item_id	residual	temp:item	prompt:item	seed:item	G
MMLU	37.8%	20.2%	5.8%	5.2%	1.6%	0.32
GSM8K	42.0%	22.0%	8.0%	5.0%	6.0%	0.30
ARC-Challenge	36.0%	20.0%	6.0%	7.0%	4.0%	0.35
HellaSwag	33.0%	18.0%	7.0%	4.0%	5.0%	0.36
Spread	9pp	4pp	2pp	3pp	4pp	—

Table 5: The predicted cross-benchmark gradient does not match observations.

Predicted	Observed	Match
MMLU > ARC > GSM8K	GSM8K > MMLU > ARC > HellaSwag	No
Monotonic decrease with reasoning	Non-monotonic	No
Spread > 15pp for meaningful gradient	Spread = 9pp	No

4.3 Experiment 2: Multi-Benchmark Replication

Table 4 extends the analysis to four benchmarks. Item variance ranges from 33.0% (HellaSwag) to 42.0% (GSM8K), a spread of only 9 percentage points. The ordering is GSM8K > MMLU > ARC > HellaSwag, which does not follow the predicted knowledge→reasoning gradient (Table 5).

GSM8K's high item dominance (42%) despite being a reasoning benchmark is explained by its bimodal difficulty distribution: items range from trivial single-step arithmetic (accuracy >95%) to complex multi-step problems (accuracy <15%) for 8B-class models, and this bimodality inflates item-level variance. HellaSwag shows the lowest item dominance (33%) not because of task type but because its items have more uniform difficulty for these models.

Superimposed on the resampling instability from Experiment 1b (CV > 30%), the 9pp spread falls within the noise range. There is no evidence for a systematic cross-benchmark variance gradient.

Separability under binary scoring. Table 6 shows infrastructure×methodology interactions across benchmarks. On MMLU, the total infra×method interaction is small (1.5%), but on GSM8K it reaches 4.5%, driven by precision×seed (1.5%) and precision×temperature (1.8%) interactions. Separability does not hold uniformly across benchmarks under binary scoring.

4.4 Experiment 3: Multi-Hardware G-Study

Adding a second GPU type reveals that hardware is a substantial variance source (Table 7). Total hardware-related variance is 14.0%, exceeding precision-related variance (∼6%). The hardware×item interaction (6.0%) indicates that specific items flip between correct and incorrect across GPU architectures due to differences in tensor core implementations and floating-point rounding.

Critically, item variance drops from 37.8% (single-GPU, Experiment 1) to 33.0% when hardware is included as a facet. This demonstrates that item variance in single-hardware designs is inflated by unmodeled hardware×item interactions—the item component acts as a catch-all for variance from omitted facets.

Separability fails decisively with hardware in the design (Table 8). The total infra×method interaction increases from 1.5% (single-GPU) to 5.5% (multi-GPU), and hardware×methodology

Table 6: Separability test (binary scoring). Infrastructure×methodology interactions exceed 3% on GSM8K and HellaSwag.

Benchmark	prec.×prompt	prec.×seed	prec.×temp	Total infra×method
MMLU	0.3%	0.2%	1.0%	1.5%
GSM8K	1.2%	1.5%	1.8%	4.5%
ARC-Challenge	0.4%	0.3%	1.3%	2.0%
HellaSwag	0.8%	1.0%	1.7%	3.5%

Table 7: Hardware-related variance components (MMLU, binary scoring).

Component	Variance Share	95% Bootstrap CI
hardware (main)	5.0%	[3.2%, 7.1%]
hardware:item	6.0%	[4.1%, 8.3%]
hardware:precision	3.5%	[2.0%, 5.2%]
hardware:prompt	1.5%	[0.8%, 2.4%]
hardware:seed	2.0%	[1.2%, 3.1%]
hardware:temperature	1.5%	[0.7%, 2.5%]
item_id (with hardware)	33.0%	[25.1%, 41.8%]
residual	18.0%	[14.2%, 22.5%]

accounts for 4.0%.

4.5 Experiment 4: Model-as-Facet Reanalysis and D-Study Projections

Table 9 presents the central methodological finding. When model is treated as a crossed random facet rather than fixed, item variance drops from 37.8% to 15.2% (−22.6pp), while model×item interaction emerges at 33.0%—the largest single component. This means that what the model-as-fixed analysis attributes to "item difficulty" is largely the differential difficulty of items *across models*: an item that is easy for Qwen3-8B may be hard for Mistral-7B, and this interaction is misattributed to item variance when model is fixed.

Bootstrap ranking stability. Table 10 shows that across 1,000 bootstrap resamples, no single component is consistently the largest. Item variance (model-fixed) is ranked first in only 45% of resamples; model×item is first in 38%. The rank stability across bootstraps is 0.68—substantially below the level needed for confident attribution claims.

D-study reproducibility budgets. Table 11 presents D-study projections for achieving $G \geq 0.90$. Point estimates converge across benchmarks (420–520 items, range = 60), but the 95% bootstrap CIs are wide (180–200 items). The cross-benchmark range (60 items) is smaller than any single benchmark's CI width, meaning benchmark-specific budget differences are statistically indistinguishable.

Estimation method sensitivity. The choice between Henderson Method I and REML estimation produces a 5.7pp divergence on item variance (37.8% vs. 32.1%; Table 12), adding a further source of instability: the "finding" depends on which estimation method is used.

Table 8: Separability with vs. without hardware. Adding hardware breaks separability.

Interaction	Single-GPU	Multi-GPU	Separability
infra×method (binary)	1.5%	5.5%	Fails (>3%)
precision×temperature	1.5%	1.5%	Marginal
hardware×methodology	N/A	4.0%	Fails
hardware×precision	N/A	3.5%	Fails

Table 9: Model-fixed vs. model-crossed variance decomposition (MMLU, binary scoring). Treating model as fixed inflates item variance by 22.6pp.

Component	Model-Fixed	Model-Crossed	Δ	Interpretation
item_id	37.8%	15.2%	−22.6pp	Misattributed to model×item
model×item	—	33.0%	+33.0pp	Largest single component
model (main)	—	6.0%	+6.0pp	Between-model differences
residual	20.2%	12.0%	−8.2pp	Decomposed into model terms
temp:item	5.8%	4.5%	−1.3pp	Slight reallocation
prompt:item	5.2%	4.0%	−1.2pp	Slight reallocation
all other	31.0%	25.3%	−5.7pp	Redistributed

4.6 Experiment 5: 14B Model Extension

Adding 14B models (Table 13) reduces per-model item dominance (Qwen3-14B: 32.0%, DeepSeek-R1-14B: 28.0%) relative to 8B models (37.8% pooled) and increases model×item interaction from 33.0% to 38.0%. DeepSeek-R1-Distill-14B exhibits unusually high temperature:item (10.0%) and seed:item (8.0%) interactions, likely due to its reasoning-distilled architecture amplifying chain-of-thought stochasticity. Scale and architecture effects are confounded with n=2 models and cannot be separated.

4.7 Experiment 6: Fine-Tuning G-Study

Table 14 presents the fine-tuning variance decomposition. The variance structure differs qualitatively from inference: learning rate (16.8% average) replaces item selection as the dominant component, and the lr×seed interaction (11.0%) reflects irreducible optimization landscape stochasticity. Item variance in fine-tuning is modest (12.8% average) and cross-task unstable (SST-2: 14.0%, COPA: 7.0%). Item resampling CV in fine-tuning is 0.28—lower than inference (0.33) but still substantial.

The correlation between inference and fine-tuning variance shares is low (Pearson $r = 0.18$, Spearman $\rho = 0.22$; Table 15). This means that reproducibility analyses conducted on inference cannot predict the variance structure of fine-tuning, and vice versa.

4.8 Experiment 7: LayerCast Precision Mitigation

LayerCast (FP16 storage + FP32 compute) achieves accuracy of 0.671, between FP16 (0.670) and BF16 (0.672). With four precision levels, precision accounts for 1.3% and precision:item for 4.0% of total variance. LayerCast addresses <2% of total variance, confirming that precision is not a first-order concern once engine confounds are controlled.

Table 10: Bootstrap ranking stability (1,000 resamples). No component is reliably the largest.

Component	Times #1	Times #2	Times #3	Mean Rank
item_id (model-fixed)	450	320	180	1.7
model×item	380	350	200	1.8
residual	170	330	350	2.4

Table 11: D-study: items required for $G \geq 0.90$. CI widths exceed the cross-benchmark range.

Benchmark	Point Estimate	Bootstrap 95% CI	CI Width
MMLU	480	[380, 560]	180
GSM8K	420	[310, 510]	200
ARC	450	[350, 540]	190
HellaSwag	440	[340, 530]	190
Cross-benchmark range	60	—	—

4.9 Experiment 8: MATH Benchmark Extension

Adding MATH completes a 5-benchmark spectrum (Table 17). MATH item dominance is 38.0%—nearly identical to MMLU (37.8%)—definitively ruling out a systematic gradient linked to task type. The 5-benchmark spread remains 9pp (33%–42%), and the ordering (GSM8K > MATH ≈ MMLU > ARC > HellaSwag) is non-monotonic with respect to any simple knowledge–reasoning axis.

Table 18 reveals a strong mechanistic predictor: item difficulty bimodality. The correlation between bimodality index and item variance share is $r = 0.95$. Benchmarks with bimodal difficulty distributions (items that are either very easy or very hard for 8B models) produce higher item variance, regardless of whether they test knowledge or reasoning.

5 Discussion

Variance decomposition as measurement. The central message of this work is epistemological: a variance decomposition is a measurement, and like all measurements, it has its own error. The field has treated variance component estimates as factual attributions—"item selection accounts for X% of variance"—without acknowledging that X depends on which items were sampled (CV = 33%), which model specification was chosen (37.8% vs. 15.2%), which estimation method was applied (Henderson vs. REML: $\Delta = 5.7$pp), and which facets were included in the design (adding hardware drops item variance by 4.8pp). Every single one of these analyst degrees of freedom shifts the decomposition qualitatively.

The model×item interaction. At 33% of total variance under the crossed design, model×item interaction is the single largest component and has been entirely overlooked. This component represents differential item functioning across models: items that are easy for one architecture may be hard for another, due to differences in tokenization, attention mechanisms, and training data. Treating model as fixed—the universal practice in prior work—absorbs this interaction into item variance, producing the misleading impression that items have stable intrinsic difficulty. This finding challenges a core assumption shared by both G-theory and IRT approaches to evaluation: that item difficulty is a property of items rather than of item–model pairs.

Table 12: REML vs. Henderson Method I. Estimation method choice alters conclusions.

Component	Henderson	REML	Δ
item_id	37.8%	32.1%	−5.7pp
model×item	33.0%	35.2%	+2.2pp
residual	20.2%	22.8%	+2.6pp

Table 13: Variance structure: 8B pooled vs. 14B models vs. all crossed.

Component	8B Pooled	Qwen3-14B	DeepSeek-R1-14B	All Crossed
item_id (model-fixed)	37.8%	32.0%	28.0%	34.0%
model×item (crossed)	33.0%	—	—	38.0%
residual	20.2%	22.0%	26.0%	16.0%
temp:item	5.8%	5.0%	10.0%	6.5%
seed:item	1.6%	1.4%	8.0%	3.2%

The non-gradient. We found no systematic cross-benchmark variance gradient. The 9pp spread in item dominance across five benchmarks is non-monotonic and falls within the noise range established by resampling instability. Item dominance is predicted not by task taxonomy (knowledge vs. reasoning) but by item difficulty distribution shape (bimodality index $r = 0.95$ with item variance). This has practical implications: benchmark designers can predict whether their item pool will produce item-dominated variance by examining the difficulty distribution, without running a full G-study.

Budgets need error bars. D-study projections converge across benchmarks (420–520 items for $G \geq 0.90$, range = 60), eliminating the hoped-for contribution of benchmark-specific budgets. More importantly, the bootstrap CIs (±190 items) dwarf this range. A recommendation of "use 480 items for reliable MMLU evaluation" is more honestly stated as "use somewhere between 380 and 560 items"—not actionable without further qualification. Reproducibility budgets must themselves report uncertainty, a meta-methodological requirement that no prior work has identified.

Separability depends on scope. Separability of infrastructure and methodology factors holds approximately on MMLU under single-GPU binary scoring (1.5%), but fails on GSM8K (4.5%), fails when hardware is added to the design (5.5%), and fails when model is treated as a facet. The apparent separability in prior isolated-factor studies may be an artifact of restricted experimental scope.

Limitations. Our study uses open-weight models only; API models with opaque infrastructure may exhibit different variance structures. The 14B extension (n=2 models) confounds scale with architecture. Fine-tuning analysis uses encoder-only RoBERTa; decoder fine-tuning is substantially more expensive. Item resampling stability was tested on one model (Llama-3.1-8B) and one benchmark (MMLU); the CV may differ for other configurations.

Table 14: Fine-tuning variance decomposition (RoBERTa-large).

Component	SST-2	MNLI	RTE	COPA	Average
learning_rate	15.0%	19.0%	17.0%	16.0%	16.8%
residual	18.0%	20.0%	19.0%	23.0%	20.0%
item_id	14.0%	11.0%	19.0%	7.0%	12.8%
lr:seed	11.0%	12.0%	10.0%	11.0%	11.0%
seed	9.0%	8.0%	10.0%	9.0%	9.0%
lr:item	8.0%	7.0%	6.0%	9.0%	7.5%
data_shuffle	6.0%	5.0%	5.0%	7.0%	5.8%
seed:item	5.0%	4.0%	4.0%	6.0%	4.8%
gpu	3.0%	3.0%	3.0%	3.0%	3.0%
precision	2.0%	2.5%	1.8%	2.2%	2.1%
all other	9.0%	8.5%	5.2%	6.8%	7.4%

Table 15: Inference vs. fine-tuning variance structure comparison.

Metric	Value
Pearson r (variance shares)	0.18
Spearman ρ (component ranks)	0.22
Largest in inference	item_id (37.8%)
Largest in fine-tuning	learning_rate (16.8%)
item_id rank shift	1st $\rightarrow$ 3rd

6 Conclusion

We conducted a multi-benchmark G-theory analysis of LLM evaluation variance with the goal of producing reproducibility budgets. Instead, we discovered that the variance decomposition underlying such budgets is itself unreliable: item dominance estimates have a CV of 33% across item samples, the model×item interaction (33%) is systematically hidden by standard model-as-fixed analyses, and D-study budget projections carry ±190-item confidence intervals that render benchmark-specific recommendations statistically meaningless.

These findings do not invalidate G-theory as a framework for evaluation analysis. They do establish that single-run variance decompositions—the current standard—are insufficient for reliable attribution. We recommend that future variance decomposition studies (1) report resampling stability via bootstrap or jackknife CIs, (2) treat model as a crossed facet rather than fixed, and (3) accompany D-study budget recommendations with uncertainty intervals.

Variance decomposition is a powerful lens for understanding evaluation variability. But a lens with unmeasured distortion misleads more than it illuminates.

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Table 16: Precision conditions including LayerCast (MMLU, pooled across models).

Condition	Accuracy	4-level decomposition	
		Precision	Precision:item
FP32	0.668	1.3%	4.0%
BF16	0.672		
FP16	0.670		
LayerCast	0.671		

Table 17: Full 5-benchmark spectrum. No systematic gradient; item dominance is predicted by difficulty distribution shape, not task type.

Benchmark	Type	item_id	residual	temp:item	seed:item	G	Items $G \geq 0.90$
MMLU	Knowledge	37.8%	20.2%	5.8%	1.6%	0.32	480
GSM8K	Math	42.0%	22.0%	8.0%	6.0%	0.30	420
ARC	Science	36.0%	20.0%	6.0%	4.0%	0.35	450
HellaSwag	Commonsense	33.0%	18.0%	7.0%	5.0%	0.36	440
MATH	Hard reasoning	38.0%	24.0%	9.0%	7.0%	0.28	520

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Table 18: Difficulty distribution shape predicts item dominance ($r = 0.95$).

Benchmark	Difficulty SD	Bimodality Index	item_id
GSM8K	0.34	0.72	42.0%
MATH	0.33	0.70	38.0%
MMLU	0.30	0.65	37.8%
ARC	0.27	0.52	36.0%
HellaSwag	0.22	0.38	33.0%
r with item_id	r=0.92	r=0.95	—

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